

Comparing Genetic Algorithms and Principal Component Analysis in Reducing Feature Dimensionality

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Abstract

It is important to do dimensionality reduction or feature selection so that machine learning models can be built in an efficient and interpretable way, especially with high-dimensional datasets. Using data up to October 2023, this study compares two methods, namely, Genetic Algorithm (GA) and Principal Component Analysis (PCA), to assess their usefulness for dimensionality reduction while preserving predictive performance. As a case study, we applied the feature selection methodology on the Breast Cancer Wisconsin (Diagnostic) dataset, comprised of 30 real-valued features that describe the characteristics of cell nuclei. PCA used to reduce dimensional space of dataset by explaining 95% of variance and GA is used to make a minimal subset of subset of relevant features based on fitness function. To evaluate the effect of the reduced dimensionality on classification accuracy, a Random Forest classifier was used. Experimental results shown that GA selected features provide accuracy from the GA by 98.25% and PCA accuracy from the PCA with 93.86% which at the cost of high computational cost. Finally, visualizations illustrated the variance retained by PCA, how features provided importance in model performance using GA, and how both influence the models. The quantitative results of this study can be used to identify the trade-off between the statistical approaches and heuristic approaches showing what needs to be prioritized in terms of application specificity when searching for dimensionality reduction methods.

Keywords: Machine Learning; Genetic Algorithms, Principal Component Analysis; Feature Selection; Dimensionality Reduction; Predictive Modeling; Reinforcement Learning

1. Introduction

Machine learning(ML) has emerged as a transformative technology that enables systems to learn from data and make decisions without explicit programming. ML techniques have been applied in a wide range of applications across domains like healthcare, finance, and autonomous systems to solve complex problems and uncover actionable insights [Virmani et al. \(2024\)](#). Among the wide variety of applications, feature selection and dimensionality reduction play a vital role in the development of efficient and interpretable ML models.

Medical diagnostics, for example, can contain many redundant or irrelevant features, as is the case with a breast cancer classification dataset. That includes very high dimensionality and introduces some challenges related to computational efficiency and model overfitting

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[Moslemi \(2023\)](#). Methods of dimensionality reduction like PCA and genetic algorithms were developed in a way to reduce the feature space while retaining information of interest.

In this work, two different approaches are compared: Principal Component Analysis (PCA) and Genetic Algorithm (GA). Principal Component Analysis is a statistical procedure in which features are transformed into orthogonal components that explain the greatest part of the data variation [Härdle et al. \(2024\)](#). On the other hand, Genetic Algorithm (GA), inspired by the principle of natural selection, directly selects an optimal subset of features [Rostami et al. \(2021\)](#). This once again underlines the trade-off between computational complexity and predictive performance.

It will hopefully provide insight into how PCA and GA perform dimension reduction and work on the diagnosis of breast cancer.

1.1. Background and Motivation

Dimensionality reduction plays a major role in ML to overcome the “curse of dimensionality.” That occurs when the datasets with more features demand exponentially increasing computational resources and run the risk of overfitting. Effective feature reduction not only improves the efficiency of ML algorithms but also enhances their interpretability with a focus on the most relevant data attributes [Ghojogh et al. \(2023\)](#).

The general tendency of medical datasets when it comes to the diagnosis of the breast cancer is to be highly dimensional with redundant features; dimension reduction is hence important for the building of various diagnostic models that perform well with good speed [Kabir et al. \(2023\)](#). The PCA and GA reflect two major classes of proposed approaches, where, in a methodical way, PCA offers a projection on low spaces based on maximized variabilities, while on the other hand, GA’s depend upon heuristic optimization to conduct a natural selection of those relevant attributes [Härdle et al. \(2024\)](#); [Rostami et al. \(2021\)](#).

1.2. Challenges

Reducing the dimensionality of high-dimensional datasets includes the following challenges: *Computational Complexity*: High-dimensional data requires huge computational power while training ML models and, hence, in applying the feature reduction techniques. Similarly, PCA involves matrix decompositions, whereas GA requires iterative optimization across the population of solutions.

Information Loss: The main drawback of reducing dimensions is the loss of some useful information. Although PCA maintains the variance, it may discard some features relevant to the domain. GA reduces this risk since it optimizes classification performance directly; this can be computationally expensive.

Interpretability: The components provided by PCA are hard to interpret; thus, it becomes difficult to relate the results to the original features. Features selected through GA provide interpretability, but their best subsets depend so much on algorithmic tuning.

1.3. Objective

The objective is to compare the performance of PCA and GA in dimensionality reduction of high-dimensional datasets while retaining predictive performance. Trade-offs between accuracy in computational complexity and interpretability for both methods. These will be

applied to a given dataset for the diagnosis of breast cancer cases and studied to observe the impact of this technique on model performance.

1.4. Contribution

The key contributions of this paper are as follows:

- A comparative study in detail on PCA and GA for the reduction of dimensionality from a high-dimensional dataset.
- Experimental results show that the proposed GA yields higher accuracy, 98.25%, than PCA, which stood at 93.86%, though at increased computational costs.
- Visualizations that give intuition on feature selection and the effect of reducing the dimensionality, including:
 1. Proportion of variance explained by PCA components.
 2. Comparison of feature selection by PCA and GA.
 3. Feature importance from Random Forest models.
- Discussion of practical trade-offs between statistical and heuristic methods of reducing dimensionality.

2. Related Work

Dimensionality reduction and feature selection are essential steps in machine learning and data preprocessing, especially when dealing with high-dimensional datasets. The goal of feature selection methods is to identify a subset of relevant features by eliminating either redundant or irrelevant features, which contributes to enhancing model performance and reduces computational complexity. Several techniques have been developed over time for these purposes, and among them, PCA and GA are two of the most studied approaches.

Since these algorithms have also become flexible and adaptable to hard optimization problems, several works investigate their efficiency in feature selection. A recent approach by [Rostami et al. \(2021\)](#) was intended to propose a community detection-based GA that first clustered the features and then performed some refinements in the selection by using a repair mechanism. The method was benchmarked with nine different classification problems outperforming Particle Swarm Optimization, Ant Colony Optimization, and Artificial Bee Colony algorithms. Therefore, the obtained results clearly showed that this approach allowed an average improvement in accuracy: 0.52%, 1.20%, and 1.57% for GAbased method versus PSO, ACO, and ABC, respectively. So it is efficient feature dimensionality reduction with retained predictive capability.

In network security, feature selection problems with common high-dimensional data have been addressed with various enhanced techniques using GA. One study by [Halim et al. \(2021\)](#) introduced the use of the GA-based Feature Selection technique for optimization in intrusion detection systems. Employing novel fitness functions and parameter tuning, this approach was tested on three benchmark datasets: CIRA-CIC-DOHBrw-2020, UNSW-NB15, and Bot-IoT. Results have shown that the proposed method significantly improved

the classification accuracy to a maximum of 99.80%, outperforming traditional methods of feature selection. The work highlights the flexibility of GA for domain-specific requirements and the performance optimization of models in high-dimensional spaces.

Principal Component Analysis has been one of the basic techniques of dimensionality reduction across a wide range of different fields. That is one of the reasons it may be preferred for many applications since it can transform data into orthogonal components that capture the most significant variance. PCA has also been used in medical diagnosis, such as reducing the dimensionality of image data for DR detection. Herein, the authors [Biswas and Banik \(2024\)](#) applied PCA to feature extraction from color fundus photographs and then went further with deep learning classification using transfer learning with CNNs. The approach achieved accuracies of 93.67%, 91.94%, and 94.40% with ResNet50, SqueezeNet1, and ResNet152, respectively. These results demonstrated the practicality of PCA in clinical applications where the reduction of dimensionality is of prime importance to handle large-scale datasets and improve diagnostic accuracy [3].

While PCA is a very popularly adopted method nowadays, there is an inherent limitation, especially with the interpretability of this approach since it is a linear method and thus can't capture complex interactions among data. To overcome these drawbacks, a sophisticated versions of PCA have been proposed by [Cottrell et al. \(2023\)](#). the Persistent Laplacian-enhanced PCA introduces spectral graph theory to analyze the data in multiple scales and to capture higher-order interactions among data. Applied to micro-array datasets for cancer classification problems, PLPCA outperformed traditional PCA and other state-of-the-art methods in bringing out intricate patterns that helped improve the classification performance.

The vastness of the number of analyzed genetic variants requires necessary dimensional reduction in genomic studies. Among them, a couple of popular approach by [Kaufman et al. \(2024\)](#) combines PCA with deep neural networks: DROP-DEEP aims for the optimal PRS prediction model. First, unsupervised PCA is used to reduce the dimensionality of the SNV datasets, and then it trains the prediction models in a supervised manner. In this regard, applied to the UK Biobank dataset, DROP-DEEP outperforms current state-of-the-art PRS methodologies that rely on PCA for data dimensionality reduction with sophisticated machine learning model strengths in processing high-dimensional genetic data.

Other works compare the performance of meta-heuristic algorithms with those from statistical methods. The work by [Isa et al. \(2024\)](#), tried a combination of PCA with the metaheuristic techniques of PSO and Genetic Algorithms to select reduced features. The study established that PCA gives a holistic view in reducing dimensions, based on variance, whereas meta-heuristic algorithms feature a fine-grained selection and perform better if combined with metaheuristics.

Particularly the feature selection methods like PCA and GA hold much medical diagnostic promise. In recent pneumonia detection using chest X-rays, both PCA and GA showed their prowess in feature selection. A proposed system by [Abdullah et al. \(2023\)](#) for a hybrid PCA-GA approach achieved an accuracy as high as 97.44%, way above when compared to individual methods. This work is focused on how statistical and heuristic methods can be combined in such a way that may efficiently reduce the workload of classification tasks within medical imaging, where expert interpretation of raw data is cumbersome and time-consuming.

PCA and GA have played a huge role in security when it comes to IoT, where high-dimensional data from different sensors are involved, which also brings challenges with them. One study by [Mohy-Eddine et al. \(2023\)](#) developed a network intrusion detection system using PCA and GA separately for feature selection. The model showed an accuracy of 99.99% on the Bot-IoT dataset, reducing the prediction time from over 51,000 seconds to less than a minute after feature selection. This result emphasizes how important dimensionality reduction is to enhance both accuracy and computational efficiency, which are essential for real-time IoT security applications. Another work by [Mohy-eddine et al. \(2023\)](#), related to intrusion detection in IoT, used PCA, GA, and a univariate statistical test for feature selection. It also provides very high accuracy, while the training time decreases substantially. By choosing those features identified by different selection methods, the model reduced the dataset to five critical features, while its performance remained high, reaching 97% for the Matthews Correlation Coefficient metric [3]. These studies collectively underline how PCA and GA can be modified or adapted to optimize the feature subset with respect to complexity presented by imbalanced, high-dimensional IoT datasets.

Apart from network security, feature selection has also contributed significantly in the field of cybersecurity. The study by [Ariawan et al. \(2024\)](#), highlights that detection of malicious URLs is such an important task that prevents monetary loss and user data from cyber-attacks. Traditionally, it was done through blacklisting, which generally proves to be inefficient in the case of newly established threats. One such study combined the use of kernel PCA, SVM, and GA to develop a robust detection system. This indeed combines the two techniques, with performance improved compared to the models of SVM and GA, since the proposed technique gave an average performance of 93.52%. Its success underlines a potential synergy between the statistical transformation of PCA and optimization capabilities of GA on complex challenges in cybersecurity.

3. Dataset and Preprocessing

3.1. Dataset Description

The data [Kaggle \(n.d.\)](#) used in this paper are the Breast Cancer Wisconsin (Diagnostic) dataset, a widely used benchmark dataset to test the performance of machine learning models in medical diagnostics. The dataset consists of 569 samples, each of which is derived from a digitized image of a fine needle aspirate (FNA) of a breast mass. Each row of the dataset represents one sample described by 30 real-valued features computed from characteristics of cell nuclei in the images. These include measures of radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. While each of these features was computed as three statistics (mean, standard error, and the “worst” value - meaning the average of the three largest values), a total of 30 features was yielded. The target variable, diagnosis, consists of “M” for malicious cases and “B” for benign cases. In total, class distribution is 357 benign and 212 malignant; hence, the dataset is slightly imbalanced. There are no missing values reported in this dataset to ensure consistency in data availability within the set. This can be publicly accessed from the repository of UCI Machine Learning.

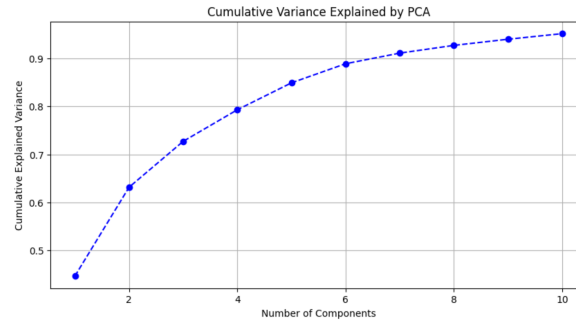


Figure 1: Commulative Variance by PCA

3.2. Data Cleaning and Transformation

The following are the steps for preprocessing this dataset, the Breast Cancer Wisconsin Diagnostic, to get it ready for analysis and model training.

Removing non-relevant features: It also drops the “id” column, which doesn’t contribute anything to the diagnosis task; thus, the dataset would contain only the relevant features.

Target Variable Encoding: The target variable “diagnosis” was converted to binary format for the machine learning algorithms. Malignant cases were encoded as 1 and benign cases as 0.

Train-Test Split: This was then split into training and test subsets using an 80-20 split. Stratification was used here to keep the original class distribution in both subsets.

Standardization: The feature scales differ; hence, ‘StandardScaler’ was used to standardize the data so that all the features are at mean zero and a standard deviation of one so each can contribute equally in training.

Handling Missing Values: Even though the data did not have missing values, mean imputation is performed as a precaution to make sure that any further pre-processing missing values would get replaced by the mean of the respective feature in order for the data to remain consistent.

These steps ensured that the dataset was clean, well-structured, and suitable for further analysis, including feature selection and model evaluation.

4. Methodology

4.1. Principal Component Analysis (PCA)

Principal Component Analysis or PCA is a technique of reducing the dimensionality in which high-order features are transformed into lower-order features while retaining a higher degree of variability. To transform the original features, it uses a new orthogonal component where the components from that will be uncorrelated and ordered according to their explaining power of variance in that data. The first principal component captures the most variance; the second captures the next largest amount of variance, and so on, each successive component being orthogonal to all the others.

First, standardization of the dataset should be done to make all features have equal importance in the analysis. A covariance matrix will then be computed to measure rela-

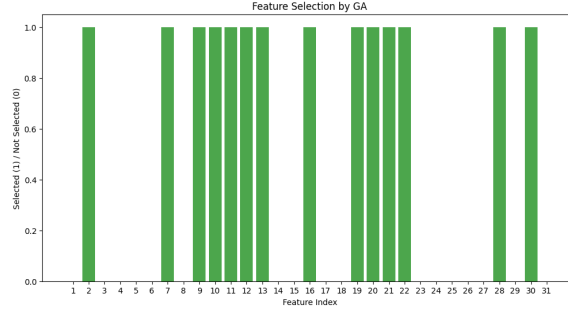


Figure 2: Features Selected by GA

tionships and variances among features. From this matrix comes out a set of eigenvectors usually called the principal components, while eigenvalues indicate the variability explained by each principal component. These are those axes in the new feature space where the size of the eigenvalues signifies their importance. Thus, PCA goes ahead selecting a subset of such components with the help of any desired criteria, like retaining only a percentage of the total variance-say 95%. This reduced number of features is simplifying the dataset where most crucial traits have remained intact. In this case, PCA was applied to the standardized dataset in order to reduce the dimensions while preserving about 95% of the variance. In such a way, the transformation will allow quick training of machine learning algorithms without a substantial loss of information. The figure 1 shows the cumulative variance explained by PCA, indicating how much variance is retained as the number of principal components increases.

4.2. Genetic Algorithm (GA)

Genetic Algorithms are a family of heuristic optimization techniques inspired by the principles of natural selection and genetics. They are especially appropriate for feature selection tasks where one wants to choose the most relevant subset of features that maximizes model performance. In this study, therefore, GA was employed to choose an optimal subset of features for classification with the view to improving predictive accuracy while keeping the number of features as few as possible.

GAs work by continuously evolving this population of candidate solutions, where each of the solutions represents one feature subset and is encoded as one binary vector. The proposed algorithm uses a developed fitness function that evaluates the fitness of all the solutions; the classification accuracy of a model trained in the selected features. With higher fitness scores, candidates enter the selection to form a new generation. Genetic operators, such as crossover and mutation, are used for the introduction of variety in such a way that an exploration in the space of solutions can be iteratively made to combine and modify the features selected by a new generation of candidate solutions As in the figure 2 shows the features selected by Genetic Algorithm.

In this case, the fitness function was related to the performance of a Random Forest classifier on the selected features. The algorithm was able to identify a minimal, informative subset of features; this reduced drastically the feature space while the accuracy preserved

or improved compared with the full dataset. Of course, this can nicely be highlighted by visualizing the progress of the feature selection, or at least by color-coding the selected features.

4.3. Classifier Used

To evaluate the performance of PCA and GA, Random Forest was used as the primary classification algorithm. Random Forest is an ensemble learning method that constructs multiple decision trees and combines their predictions to improve accuracy and robustness. It is particularly well-suited for high-dimensional data and can handle datasets with a large number of irrelevant or redundant features.

For PCA-transformed data, the Random Forest classifier was trained on the reduced set of principal components, leveraging the lower-dimensional representation to achieve efficient and accurate classification. Similarly, the classifier was trained on the subset of features selected by GA to evaluate its effectiveness in preserving essential information. Random Forest's ability to provide feature importance measures also offered insights into the relevance of selected features, further validating the outcomes of the dimensionality reduction and feature selection processes.

By using Random Forest across both PCA and GA workflows, the study ensured consistency in evaluation, enabling a direct comparison of the two methods based on their impact on classification accuracy.

5. Experiments and Results

5.1. Setup

The experiments were conducted using Google Colab, a cloud-based platform that provides free access to computational resources, including GPU and TPU support, which are highly advantageous for machine learning tasks. The dataset and supporting files were stored in Google Drive, allowing seamless integration with Colab for data access and management.

The Breast Cancer Wisconsin (Diagnostic) dataset, a well-known benchmark for classification tasks, was loaded and preprocessed directly within Colab. Key libraries such as numpy, pandas, and matplotlib were employed for data manipulation and visualization. Machine learning tasks, including model training and evaluation, relied on the scikit-learn library, which provided tools for standardization, PCA, and Random Forest classification. The geneticalgorithm library was used to implement and optimize the Genetic Algorithm (GA) for feature selection.

The experiments involved the following steps:

- Loading and preprocessing the dataset by removing irrelevant columns, encoding the target variable, and standardizing features.
- Splitting the data into training and testing sets with an 80-20 split, ensuring stratification to maintain class balance.
- Applying PCA for dimensionality reduction to retain 95% of the variance and subsequently transforming the data.

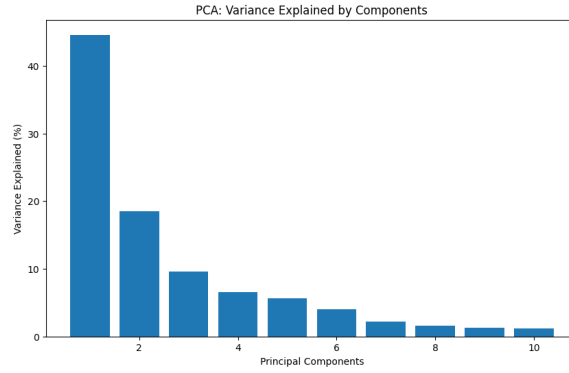


Figure 3: Explained Variance by Components

- Implementing GA to optimize feature selection, with a fitness function based on classification accuracy using the Random Forest model.
- Training and testing the Random Forest classifier on both PCA-reduced and GA-selected feature subsets.
- Recording and comparing results for accuracy, number of selected features, and computational efficiency.

The experiments were executed in an iterative manner, with outputs periodically saved to Google Drive for reproducibility and further analysis.

5.2. Results

The comparative analysis of Principal Component Analysis (PCA) and Genetic Algorithm (GA) provided valuable insights into their performance for dimensionality reduction and feature selection. Both methods significantly reduced the dimensionality of the dataset, facilitating efficient model training and evaluation.

PCA Results PCA reduced the dataset to 10 principal components, capturing 95% of the total variance. By transforming the features into a lower-dimensional orthogonal space, PCA preserved most of the dataset's information while removing redundancies and correlations. A Random Forest classifier trained on the PCA-transformed features achieved a classification accuracy of 93.86% on the testing set.

The figure 3 of the PCA components revealed that the first few components accounted for the majority of the variance, with diminishing returns for additional components. This outcome validated the selection of 10 components as an optimal balance between dimensionality reduction and information retention.

GA Results The GA approach directly optimized feature selection to maximize classification accuracy. Starting with an initial population of random feature subsets, the algorithm iteratively evolved through selection, crossover, and mutation operations. The fitness function, based on the accuracy of a Random Forest classifier, guided the optimization process.

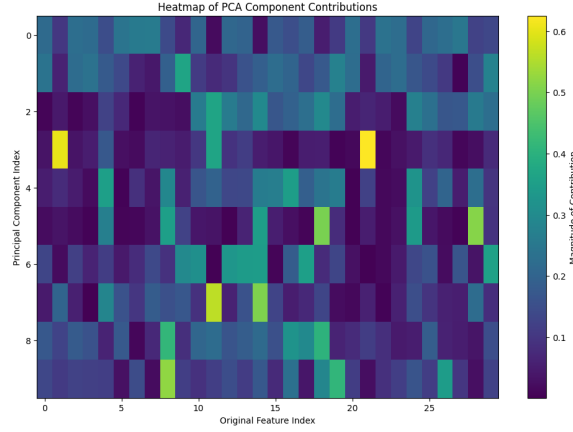


Figure 4: HeatMap

GA selected 14 features out of the original 30, representing a substantial reduction in dimensionality while retaining features most relevant to the classification task. A Random Forest classifier trained on the GA-selected features achieved a higher accuracy of 98.25% on the testing set. The improved accuracy demonstrated GA's ability to identify domain-specific features that enhanced model performance.

Comparative Analysis

Dimensionality Reduction: PCA reduced the feature space to 10 components, while GA selected 14 individual features. Although PCA achieved a greater reduction, GA provided higher accuracy by retaining domain-specific information.

Classification Accuracy: The accuracy of the Random Forest classifier was 93.86% with PCA-transformed features and 98.25% with GA-selected features. This 4.39% improvement highlights GA's strength in optimizing feature subsets for predictive tasks.

Computational Efficiency: PCA was computationally efficient, requiring a single pass through the dataset to compute the covariance matrix and perform eigen decomposition. In contrast, GA involved multiple iterations and required more computational resources due to its evolutionary operations and repeated model evaluations.

Interpretability: The principal components generated by PCA are linear combinations of original features, making them less interpretable in terms of domain relevance. Conversely, GA retains original features, providing better interpretability for domain experts.

Visualization Insights

The figure 4 illustrates the contribution of each original feature to the principal components derived by PCA. The intensity of the color indicates the magnitude of a feature's contribution, with brighter colors signifying stronger contributions. This visualization highlights how PCA transforms the original feature set into a reduced set of uncorrelated components, with certain features contributing more significantly to specific components. It provides an understanding of how variance is distributed among features and components, supporting the decision to retain components explaining 95% of the variance.

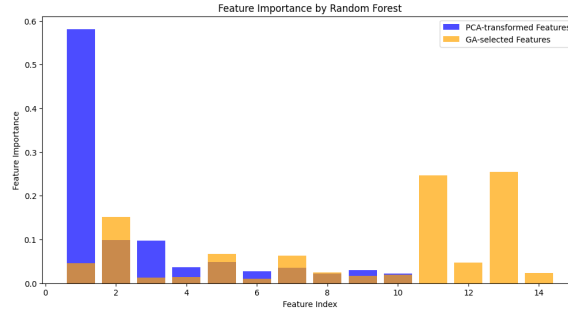


Figure 5: Feature Importance

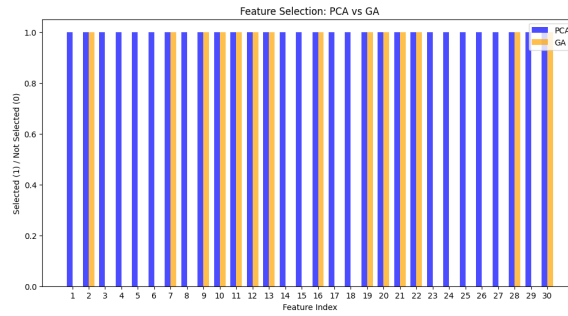


Figure 6: Feature Selection: PCA vs GA

This bar chart in figure 5 compares the feature importance scores as determined by the Random Forest classifier for PCA-transformed features and GA-selected features. The blue bars represent the importance scores for PCA components, while the yellow bars correspond to GA-selected features. The chart clearly shows that the GA-selected features retain more domain-specific relevance, as they correspond to the original features directly. In contrast, PCA components, being linear combinations of features, distribute their importance across multiple components. This visualization emphasizes the interpretability advantage of GA-selected features in comparison to PCA.

This binary indicator chart in figure 6 displays the features selected by PCA and GA. For PCA, all original features contribute indirectly to the principal components, but only a reduced set of components is retained for analysis. GA, on the other hand, selects a specific subset of original features, directly excluding irrelevant or redundant ones. This side-by-side comparison underscores GA's ability to select a smaller, more interpretable subset of features while maintaining or improving classification accuracy.

The bar chart in figure 7 comparing classification accuracies demonstrates the effectiveness of PCA and GA in dimensionality reduction. The Random Forest classifier achieved an accuracy of 93.86% when trained on PCA-transformed features and 98.25% when trained on GA-selected features. This visualization underscores the superior performance of GA in retaining domain-specific features that enhance predictive accuracy.

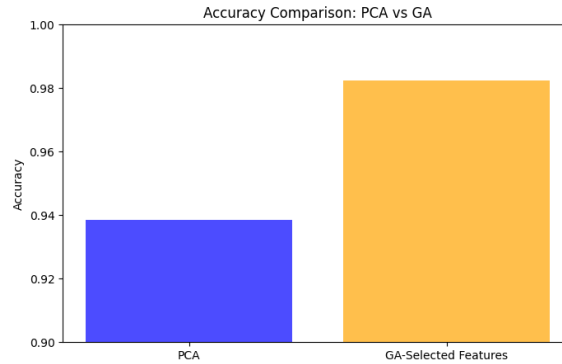


Figure 7: Accuracy Comparison: PCA vs GA

6. Conclusion

This study conducted a detailed comparative analysis of Principal Component Analysis (PCA) and Genetic Algorithm (GA) for dimensionality reduction and feature selection, using the Breast Cancer Wisconsin (Diagnostic) dataset as a benchmark. PCA effectively reduced dimensionality by transforming the original features into orthogonal principal components that captured 95% of the variance, demonstrating its strength in computational efficiency and variance preservation. However, PCA's reliance on linear transformations and lack of interpretability limited its ability to directly optimize classification performance.

On the other hand, GA provided a tailored feature selection approach, identifying a subset of 14 relevant features that maximized classification accuracy. Despite its computational overhead, GA demonstrated superior predictive performance, achieving an accuracy of 98.25% compared to PCA's 93.86%. The GA approach also preserved domain-specific interpretability by directly optimizing the feature subset for the classification task.

The findings underscore the complementary strengths of PCA and GA. PCA is an efficient choice for tasks where variance retention and computational simplicity are paramount, while GA is better suited for applications requiring domain-specific feature optimization and higher predictive accuracy. Future work could explore hybrid approaches that combine the strengths of PCA and GA or extend the analysis to other datasets and classifiers. This study provides valuable insights for practitioners seeking effective dimensionality reduction techniques tailored to specific machine learning tasks.

7. References

References

- Salma Hameedi Abdullah, Wafaa M Salih Abedi, and Raghad Mohammed Hadi. Enhanced feature selection algorithm for pneumonia detection. *Periodicals of Engineering and Natural Sciences (PEN)*, 10(6):168–180, 2023.
- Sandy Ariawan, Arvind Kumar, Chinthala Kumara Swamy, V Divya, V Manikandan, and S Rukmani Devi. Intelligent malicious url detection using kernel pca-svm-ga model with

- feature analysis. In *2024 International Conference on Data Science and Network Security (ICDSNS)*, pages 1–6. IEEE, 2024.
- Ankur Biswas and Rita Banik. Detection and classification of diabetic retinopathy based on ensemble learning. *Advances in Computational Intelligence*, 4(3):9, 2024.
- Sean Cottrell, Rui Wang, and Guo-Wei Wei. Plpca: persistent laplacian-enhanced pca for microarray data analysis. *Journal of chemical information and modeling*, 64(7):2405–2420, 2023.
- Benyamin Ghojogh, Mark Crowley, Fakhri Karray, and Ali Ghodsi. *Elements of dimensionality reduction and manifold learning*. Springer, 2023.
- Zahid Halim, Muhammad Nadeem Yousaf, Muhammad Waqas, Muhammad Sulaiman, Ghulam Abbas, Masroor Hussain, Iftekhhar Ahmad, and Muhammad Hanif. An effective genetic algorithm-based feature selection method for intrusion detection systems. *Computers & Security*, 110:102448, 2021.
- Wolfgang Karl Härdle, Léopold Simar, and Matthias R Fengler. Principal component analysis. In *Applied multivariate statistical analysis*, pages 309–345. Springer, 2024.
- Faizatulhaida Md Isa, Wan Nor Munirah Ariffin, Muhammad Shahar Jusoh, and Erni Puspantastari Putri. A review of genetic algorithm: Operations and applications. *Journal of Advanced Research in Applied Sciences and Engineering Technology*, 40(1):1–34, 2024.
- Md Faisal Kabir, Tianjie Chen, and Simone A Ludwig. A performance analysis of dimensionality reduction algorithms in machine learning models for cancer prediction. *Healthcare Analytics*, 3:100125, 2023.
- Kaggle. Breast cancer wisconsin (diagnostic). <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>, n.d. Retrieved from <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>.
- Hadasa Kaufman, Yanden Hochenberg, Michal Linial, and Nadav Rappoport. Drop-deep: Dimensionality reduction for polygenic risk score using deep learning approach. *medRxiv*, pages 2024–05, 2024.
- Mouaad Mohy-Eddine, Azidine Guezzaz, Said Benkirane, and Mourade Azrour. An efficient network intrusion detection model for iot security using k-nn classifier and feature selection. *Multimedia Tools and Applications*, 82(15):23615–23633, 2023.
- Mouaad Mohy-eddine, Azidine Guezzaz, Said Benkirane, and Mourade Azrour. An intrusion detection model using election-based feature selection and k-nn. *Microprocessors and Microsystems*, page 104966, 2023.
- Amir Moslemi. A tutorial-based survey on feature selection: Recent advancements on feature selection. *Engineering Applications of Artificial Intelligence*, 126:107136, 2023.
- Mehrdad Rostami, Kamal Berahmand, and Saman Forouzandeh. A novel community detection based genetic algorithm for feature selection. *Journal of Big Data*, 8(1):2, 2021.

Deepali Virmani, Md Atheeq Sultan Ghor, Nishant Tyagi, RP Ambilwade, Priyanka Rajesh Patil, and MK Sharma. Machine learning: The driving force behind intelligent systems and predictive analytics. In *2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies*, pages 1–6. IEEE, 2024.

Appendix A. First Appendix

This is the first appendix.

Appendix B. Second Appendix

This is the second appendix.